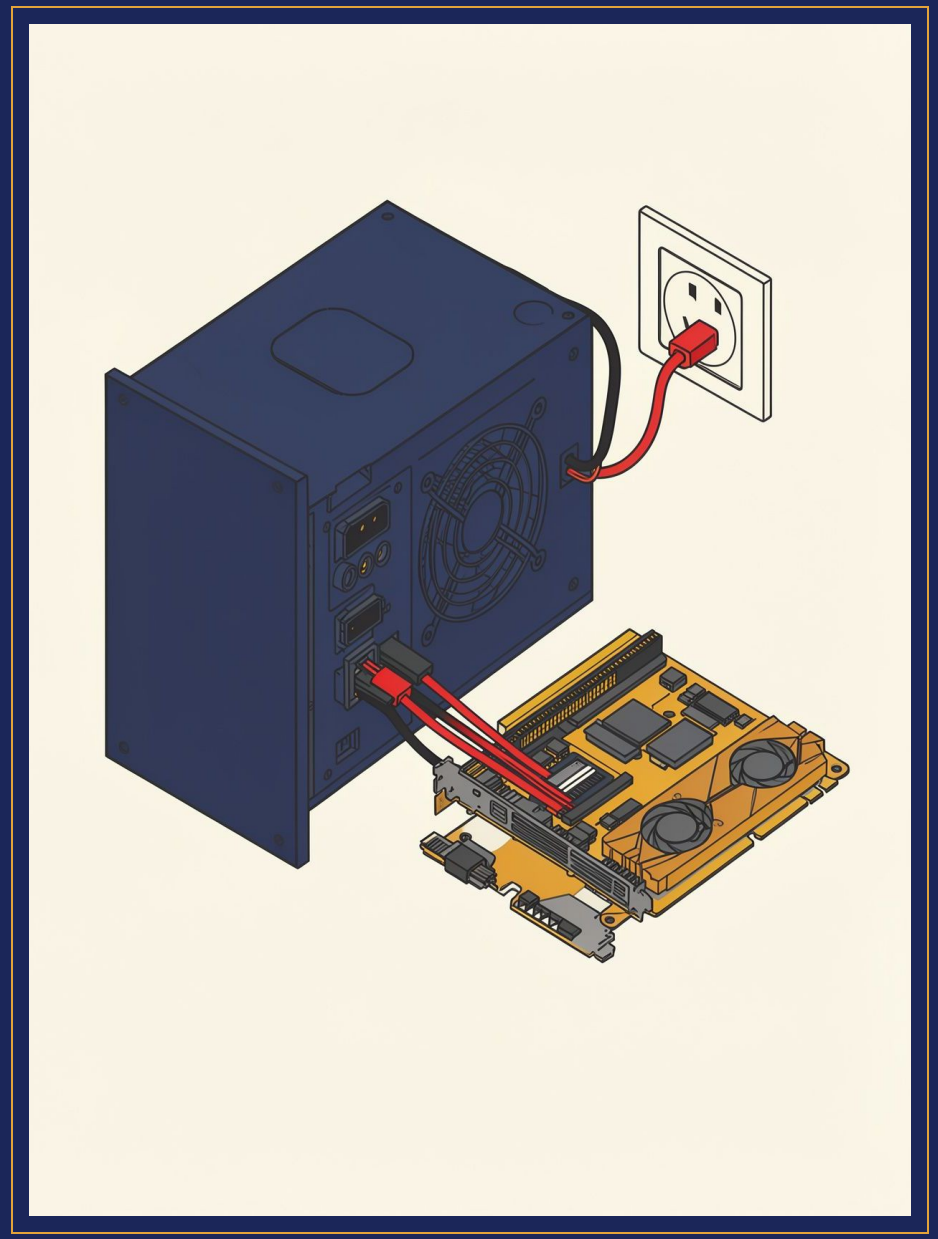
POWER SYSTEMS · EVERYDAY COMPUTING

Power supplies behind everyday *computing*

From wall power to the DC rails inside your devices.



College electronics · Core concepts and safe PC wiring

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POWER SYSTEMS

Everyday computing

TWO JOBS

SMPS changes power; UPS protects it

SMPS

Changes AC from the wall into DC for electronics.

The power supply converts and regulates the energy a device uses.

UPS

Keeps devices running during a short power cut.

The backup gives time to continue briefly or save work safely.

Core distinction

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Five devices that use SMPS or UPS

Desktop PC — SMPS
Supplies the DC voltages inside the computer.

Laptop charger — SMPS
Gives the laptop safe, regulated DC power.

Game console — SMPS
Converts wall power efficiently.

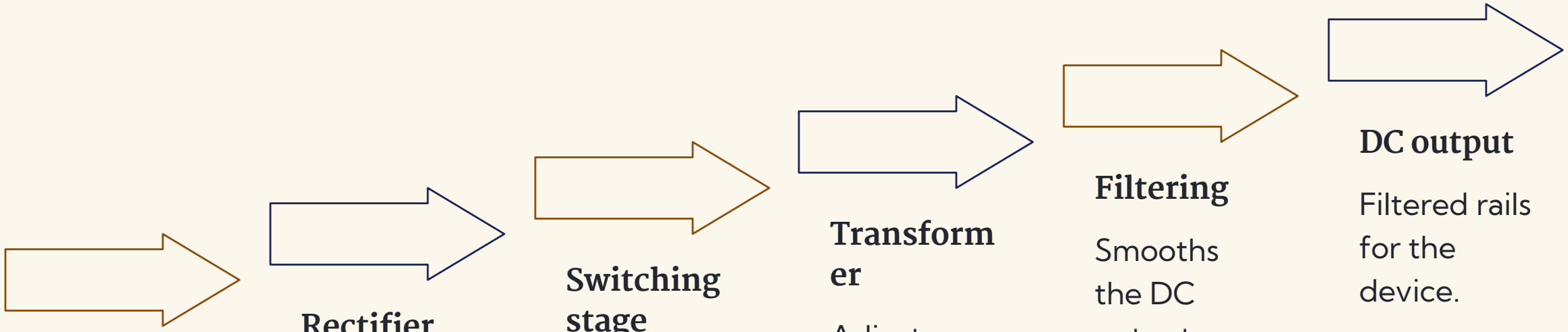
Desktop PC — UPS
Gives time to save work during an outage.

Wi-Fi router — UPS
Keeps the internet connected during a brief interruption.



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How power moves through an SMPS



AC input

Power from the wall.

Rectifier

Changes AC into pulsed DC.

Switching stage

Switches at high frequency.

Transformer

Adjusts voltage safely.

Filtering

Smooths the DC output.

DC output

Filtered rails for the device.

COMMON RAILS

+12 V | +5 V | +3.3 V

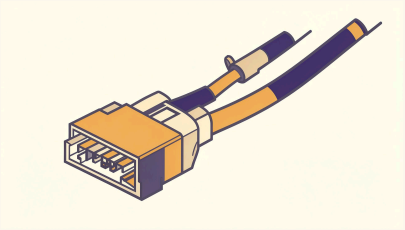
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PC HARDWARE

Everyday computing

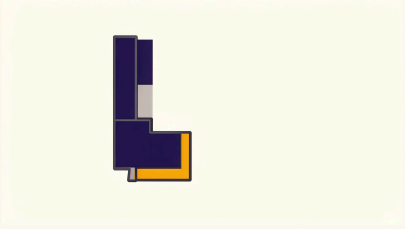
RECOGNITION

Four PC power connectors to recognize



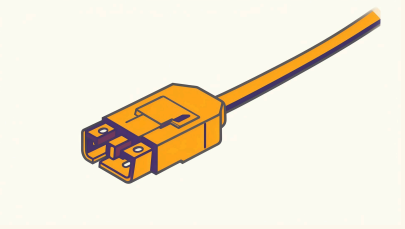
ATX 24-pin

Motherboard power



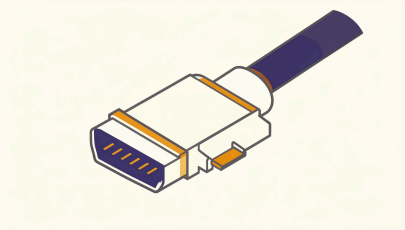
SATA power

SSD and hard-drive power



Molex

Older drives and accessories



PCIe 6/8-pin

Graphics-card power

Four connectors

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PC HARDWARE

Connector guide

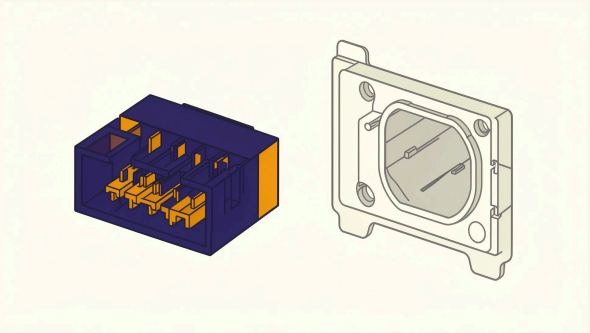
MOTHERBOARD + STORAGE

ATX 24-pin and SATA power

ATX 24-PIN

Motherboard power

Align the latch and press straight in.

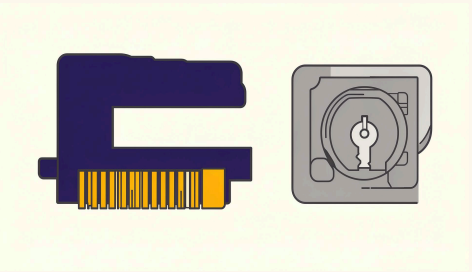


ATX 24-pin · motherboard

SATA POWER

SSD and hard-drive power

Follow the L-shaped key and pull the plastic body.



SATA power · storage

Start here

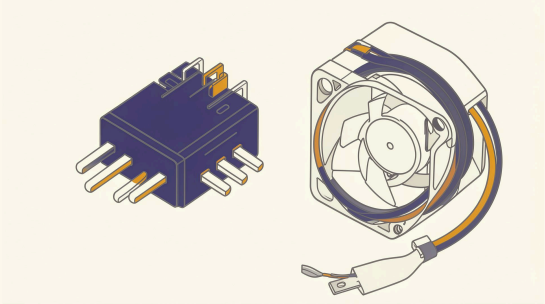
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Molex and PCIe power

MOLEX

Older drives, fans and accessories

Hold the connector, not the wires.

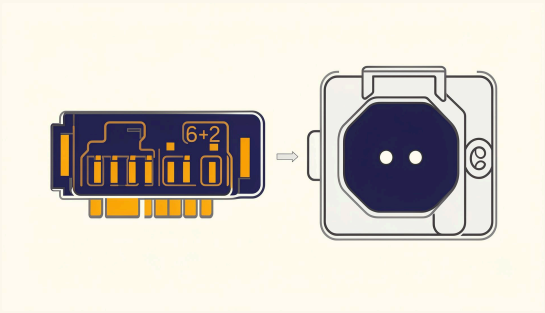


Molex · accessories

PCIE 6/8-PIN

Graphics-card power

Use the cable labeled PCIe, not CPU/EPS.



PCIe 6+2 · graphics card

Check the label

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Gaming-PC wiring: four simple steps

Power off

Switch the PSU off and unplug it.

Motherboard

Connect the 24-pin ATX cable and 4+4/8-pin CPU cable.

Graphics card

Connect the required PCIe 6/8-pin or 6+2-pin cable.

Storage

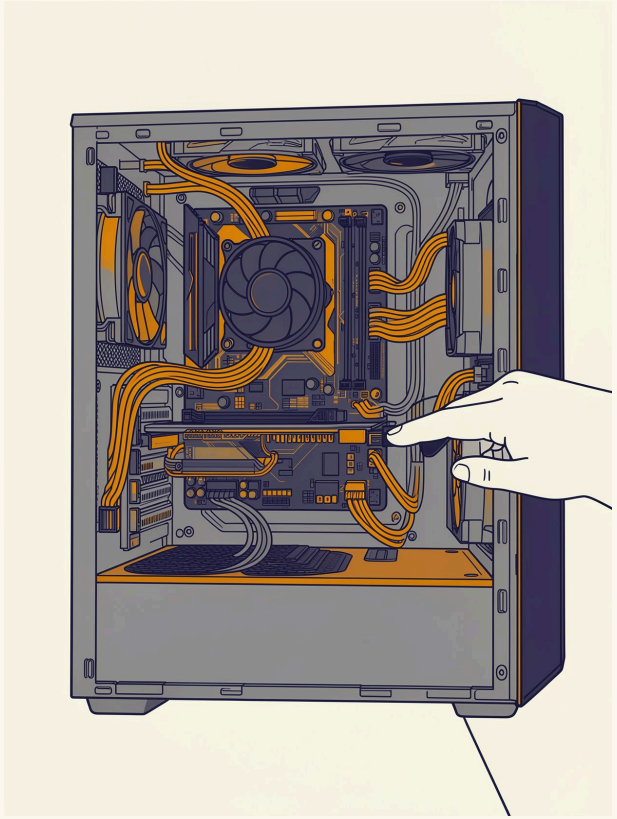
Connect SATA power to SATA drives; use Molex only for compatible accessories.

Always check the manuals. Cable needs vary by build.

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Before you press the power button

- ✓ Use only cables made for that PSU.
- ✓ Match PCIe cables to the GPU and CPU/EPS cables to the processor.
- ✓ Make sure every latch is fully seated.
- ✓ Keep cables away from fans.
- ✓ Check the manuals one last time.



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Sources

Cooler Master. *How To Safely Install a PSU?*
coolermaster.com

Intel. *How to Choose Power Supply for PC.*
intel.com

Intel. *ATX Version 3 Multi Rail Desktop Platform Power Supply Design Guide.*
Intel design guide PDF

Use the manufacturer manuals for the exact PSU, motherboard, GPU and storage devices in a build.

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